LTP 310

EOE- 076

OPEN ELECTIVES

INDUSTRIAL AUTOMATION-I

Unit 1 : Historical perspective of industrial automation
- Origin & Evolution,
- Current and future trends –
- Components of industrial automation system and their functionalities.

Unit 2 : Programmable Control Subsystem :
- Programmable Automation Controllers
- Discrete Control & Continuous Control.
- Functional Subsystems and Modules :
- Rack
- Bus
- Cables
- Power Supply
- Processor,
- IO Subsystem,
- Communications Subsystems-
- Memory Structure-
- Scanning Process.

Unit 3 : Advanced Programmable Automation Controllers:
Programmable automation controllers: Evolution & Architecture
- IT Embedded with Control Architecture
- Multidisciplinary control – Software architecture of PAC
- Programming methods
- Ladder Logic
- Structured Text (ST)
- Sequencial Function Chart (SFC)
- Functional Block Diagram (FBD)
- Tag based data structure and programming
- Asynchronous IO Update-Pre-emptive Programming Structure.

Unit 4 : VISUALIZATION : HUMAN MACHINE INTERFACE (HMI) & SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA) SYSTEMS:
- Need for HMI
- Hardware based HMI panels
- Global database : Concept and structure
- PC based HMI systems – SCADA
- Different functionalities
- Creating user screens,
- Trend, graphs and data acquisition systems – Alarm and event logging -
- User administration
- Benefits of implementing SCADA systems
- Case studies of SCADA implementation .

TEXT BOOK:

1. **Jon Stenerson, "Programmable Logic Controllers with ControlLogix"**,Delmar Publishers,2003

REFERENCES:

1. Richard L Shell and Ernest L Hall, "Hand book of Industrial Automation", Marcel Dekker Inc.,2000.
2. David Bailey and Edwin Wright, "Practical SCADA for Industry", Newness Publishers, 2003.
3. Dick Caro, "Automation Network Selection", ISA – The Instrumentation Systems and Automation Society, 2004.
4. Edwin H Zimmerman, "Getting Factory Automation Right (the first time)", Manufacturing Engineers, 2001.

INDUSTRIAL AUTOMATION-II

EOE-085
LTP 310

Unit 1 : INDUSTRIAL COMMUNICATION NETWORKS:
- Benefits of Industrial networking
- Requirements of an Industrial Network
- Concept of proprietary and open standards
- Hierarchy of networking
- Device level: Devicenet, Ethernet
- Control and Information level : ControlNet, Ethernet

Unit 2 : VFD and Servo Systems:
- Introduction to Variable Frequency Drives-(VFD)
- Industrial Variable Frequency Drives
- Parameters
- Tuning
- Motion Control Systems
- Introduction to Servo Drives and Motors
- Basic Terminology
- Distinction of application areas of VFD & Servo.

Unit 3 : Integrated Position , Velocity Control, & Predictive Diagnostics Systems:
- Motion Networks- Serial Real Time Communication Network
- ISA 1588- Precision Time Protocol(PTP)
- Using Ethenet/IP for Motion Control
- Motion as a discipline of a Programmable Automation Controller System
- Integration of VFD in an Automation environment
- Vibration Monitoring - Integrating vibration Monitoring System in an Automation System.

Unit 4 : Information Enabled Control Systems:
- Importance of Real Time Data
- Overall Equipment Effectiveness (OEE)
- Concepts& Need of measuring OEE
- Methodology & tools
- Historian:
- Need & Uniqueness of Historian Software
- Enterprise Manufacturing Intelligence (EMI)
- Reporting Tools
- Real time Web based reporting
- Asset Management
- Need of an asset management system
- Components of Asset management System
- Disaster recovery.

TEXT BOOK:

1. **Jon Stenerson, "Programmable Logic Controllers with ControlLogix"**,Delmar Publishers,2009

REFERENCES:

1. Richard L Shell and Ernest L Hall, "Hand book of Industrial Automation", Marcel Dekker Inc.,2000.
2. David Bailey and Edwin Wright, "Practical SCADA for Industry", Newness Publishers, 2003.
3. Dick Caro, "Automation Network Selection", ISA – The Instrumentation Systems and Automation Society, 2004.
4. Edwin H Zimmerman, "Getting Factory Automation Right (the first time)", Manufacturing Engineers, 2001.